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Loops

→ Loops in python are used to execute a block of code repeatedly until a certain condition is met.

Python mainly provides two types of loops

- i) for loop
- ii) while loop
- iii) nested loop

- * for loop inside for loop

- * for loop inside while loop

- * while loop inside while loop

- * while loop inside for loop

i) For loop:-

Two types:-

i) for loop with Sequence : String, list, tuple, Set, dict

ii) for loop with range

Example for for loop with sequence

1) name = "pradeep"

Print (name)

o/p:- pradeep

for loop with sequence example
name is continuous so we use
for loop:

2) name = "pradeep"
for i in name:

Print (i)

o/p:-
p
r
a
d
e
e
p

i is a iterative variable

3) for i in enumerate(name):

Print (i)

(0, 'p')

(1, 'r')

(2, 'a')

(3, 'd')

(4, 'e')

(5, 'e')

(6, 'p')

↓
this word is used to print
the position of character with
character

4) fruit = ['apple', 'mango', 'grapes', 'pineapple']

Print (fruit)

o/p:- ['apple', 'mango', 'grapes', 'pineapple']

5) for i in fruit:

Print (i)

o/p:-

apple

mango

grapes

Pineapple

other programming languages

initialisation

condition

incrementation/decrementation

In Python

range (start value, stop value, step size)

initialisation replaced by start value.

condition replaced by stop value

Note: ~~increment~~ In condition we used less than only

incrementation/decrementation replaced by step size.

Ex: For i in range (1, 11, 1):
 Print(i)

o/p: 1
2
3
4
5
6
7
8
9
10

range (1, 11, 1)

Ci = 1; i < 11; i + 1

i = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

range (1, 10, 1)

Ci = 1; i < 10; i + 1

i = 1, 2, 3, 4, 5, 6, 7, 8, 9

for i in range (1, 6, 1):

Print ("Pradeep")

o/p: Pradeep
Pradeep
Pradeep
Pradeep
Pradeep

Note: default start value is 0

default step size is 1

If u didn't mention start value & step size it will run when u only mention condition

Ex: for i in range (10):
Print (i)

o/p: 0
1
2
3
4
5
6
7
8
9

name = "Pradeep"

for i in range (0, 7, 1):

Print (i, name[i])

o/p: 0 P
1 r
2 a
3 d
4 e
5 e
6 p

Question 8:

1) Write a program to print even numbers from 1 to 20

→ For in range (2, 21, 2):
 Print (i)

o/p: 2
 4
 6
 8
 10
 12
 14
 16
 18
 20

2) Write a program to Print like this

1 = odd numbers
2 = even numbers
3 = odd numbers
⋮
10 = even numbers

~~o/p:~~ ~~1 = odd~~

code: For i in range (1, 11, 1):
 if (i % 2 == 0):
 Print (i, " = even number")
 else:
 Print (i, " = odd number")

o/p: 1 = odd numbers
 2 = even numbers
 3 = odd numbers
 4 = even numbers
 5 = odd numbers
 6 = even numbers
 7 = odd numbers
 8 = even numbers
 9 = odd numbers
 10 = even numbers